

LESSON 7-6 • SESSION 2 • PRACTICE • FORM A

Comparing Distances on the Plane

Learning target – I can decide whether to add or subtract to find a distance, then compare two distances. · 6.NS.C.8

REMEMBER

- Find **each** distance first. Then compare the two numbers.
- Changing coordinates with **opposite signs**: add. **Same sign**: subtract.
- With a **scale**, change both distances to real units before you compare. A distance is never negative.

WORKED EXAMPLE

Jo walks from $(-3, 10)$ to $(-3, 2)$. Sam walks from $(6, -8)$ to $(-7, -8)$. Each unit is 1 block. Who walks farther?

Step 1 Jo: same x , both positive, so $10 - 2 = 8$ blocks.

Step 2 Sam: same y , opposite signs, so $6 + 7 = 13$ blocks.

Answer Sam walks farther, by $13 - 8 = 5$ blocks.

WORDS distance (*distancia*) · quadrant (*cuadrante*) · compare (*comparar*)

PRACTICE

show your work · label units

1. Add or subtract? Find each distance.

a. $(5, -3)$ to $(5, -15)$ _____

b. $(5, -3)$ to $(5, 10)$ _____

c. $(-4, 1)$ to $(6, 1)$ _____

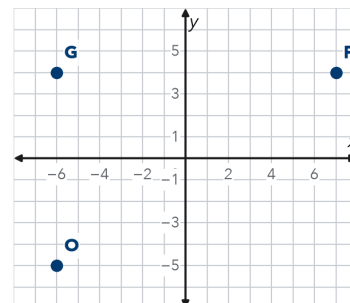
d. $(3, -8)$ to $(3, -2)$ _____

2. **Campus Map.** The map shows a gym (G), a field (F) and an office (O). Each grid unit stands for 10 meters.

a. Gym to Field _____ m

b. Gym to Office _____ m

c. Which trip is longer? By how many meters? _____



3. **Two Riders.** From $(-9, 5)$, Tariq rides to $(-9, -8)$ and Jada rides to $(6, 5)$. Each unit is 1 block. Who rides farther, and by how much?

Hint Each ride keeps one coordinate. Check the signs of the other one.

Show your work.

By _____

4. **Partner Point.** Point T is at $(-4, 16)$. Point U is 22 units directly below T. Name point U.

Hint Moving down changes only y : $16 - 22$.

Show your work.

U = _____

5. **Place the Stop.** Point N is at $(13, -5)$. Point M is 20 units directly to the left of N. Name point M.

Hint Moving left changes only x : $13 - 20$.

Show your work.

M = _____

- 6. Two Hikes.** Kendra hikes from $(-11, 12)$ to $(-11, 3)$. Owen hikes from $(-2, -15)$ to $(10, -15)$. Each unit is 1 km. Who hikes farther, and by how much?

Show your work.

By _____

- 7. Two Walks.** From $(4.25, 5.75)$, Ana walks to $(4.25, -3.25)$ and Ben walks to $(4.25, 9.75)$. Who walks farther, and by how much?

Hint Decimals work the same way. Check the signs of the y -values.

Show your work.

By _____

- 8. Delivery Routes.** A depot is at $(0, 0)$. Stop 1 is at $(0, 12)$, Stop 2 is at $(0, -9)$ and Stop 3 is at $(0, 4)$.

- Find the distance from the depot to each stop. _____
- Order the stops from closest to farthest. _____
- Find the distance from Stop 1 to Stop 2. _____ units

FIND THE ERROR

explain the mistake, then fix it

- 9. Ines's Work.** To find the distance from $(-8, 3)$ to $(4, 3)$, Ines wrote $8 - 4 = 4$ units. Explain her mistake. Then find the correct distance.

TEST PRACTICE

choose the best answer

- 10.** Two coordinates on the same side of zero call for –

A addition **B** subtraction **C** multiplication **D** division

- 11.** A distance can never be –

A zero **B** a decimal **C** negative **D** greater than 10

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

2 L	1 J	1.5 X	(33, -5) Z	3 G	21 R	(-4, 38) Q	(-4, -6) O	5 E	(-7, -5) N
			#3	#4	#5	#6	#7	#8c	

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____